

Software-based Networking: KickOff

SS2026

Michael König — m.koenig@kit.edu

Paul Seehofer — paul.seehofer@kit.edu

Institute of Telematics – Department of Informatics



“Projektpraktikum Software-basierte Netze”

- Today: Information Event
- Target audience: Students of Computer Science (Master)
- 6 ETCS

Before we start...

- Course / Discussions are done in **English**
- Documents / Reports / etc. are done in **English**

This includes

- Slides / assignments
- Presentations
- Question and answer sessions, discussions

However: Language skills are NOT graded → only content

Before we start...

The team

- Michael König (Organizer) <m.koenig@kit.edu>
- Paul Seehofer (Organizer) <paul.seehofer@kit.edu>

Agenda for Today

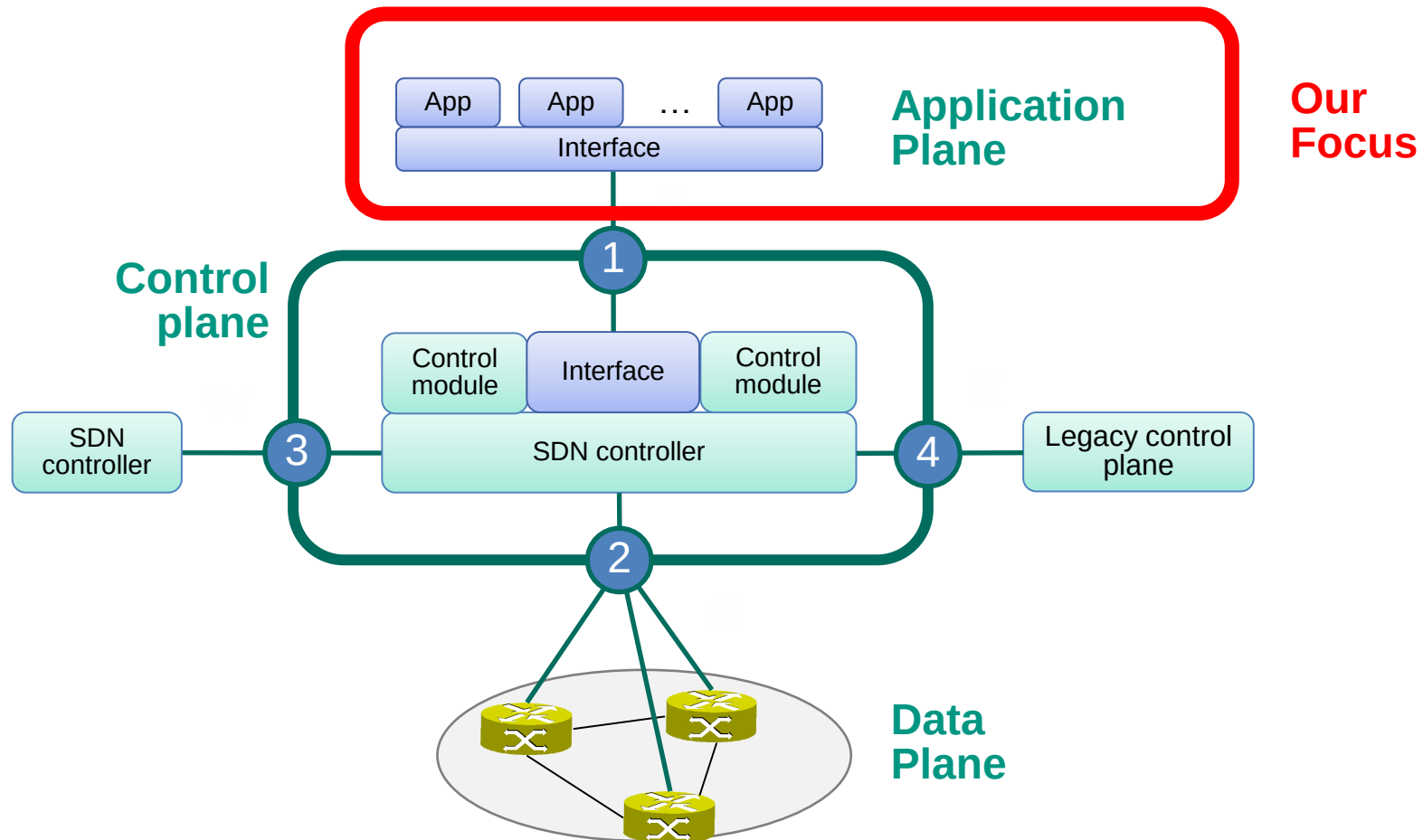
- What is this practical course about?
- Time schedule
- Overview of graded documents/activities
- Next Steps

Agenda for Today

- **What is this practical course about?**
- Time schedule
- Overview of graded documents/activities
- Next Steps

Surprise: It's about SDN ☺

- SDN = Software Defined Networking



About the course

- Organized in two blocks
 - First block: **Basics** of SDN Application Development (~6 weeks in total)
 - Second block: **Reverse-engineering** + Evaluation of an SDN application (~7 weeks in total)
- Roughly Bi-Weekly meetings
 - Starting next week
 - Wednesdays, always 15:45-17:15pm
 - Switches to a 2-week cycle in the second block
 - More details in a couple of slides...
- Important
 - Please contact us if you cannot make it to one of the meetings!
 - The earlier, the better 😊

Block1: Basics of SDN Application Development

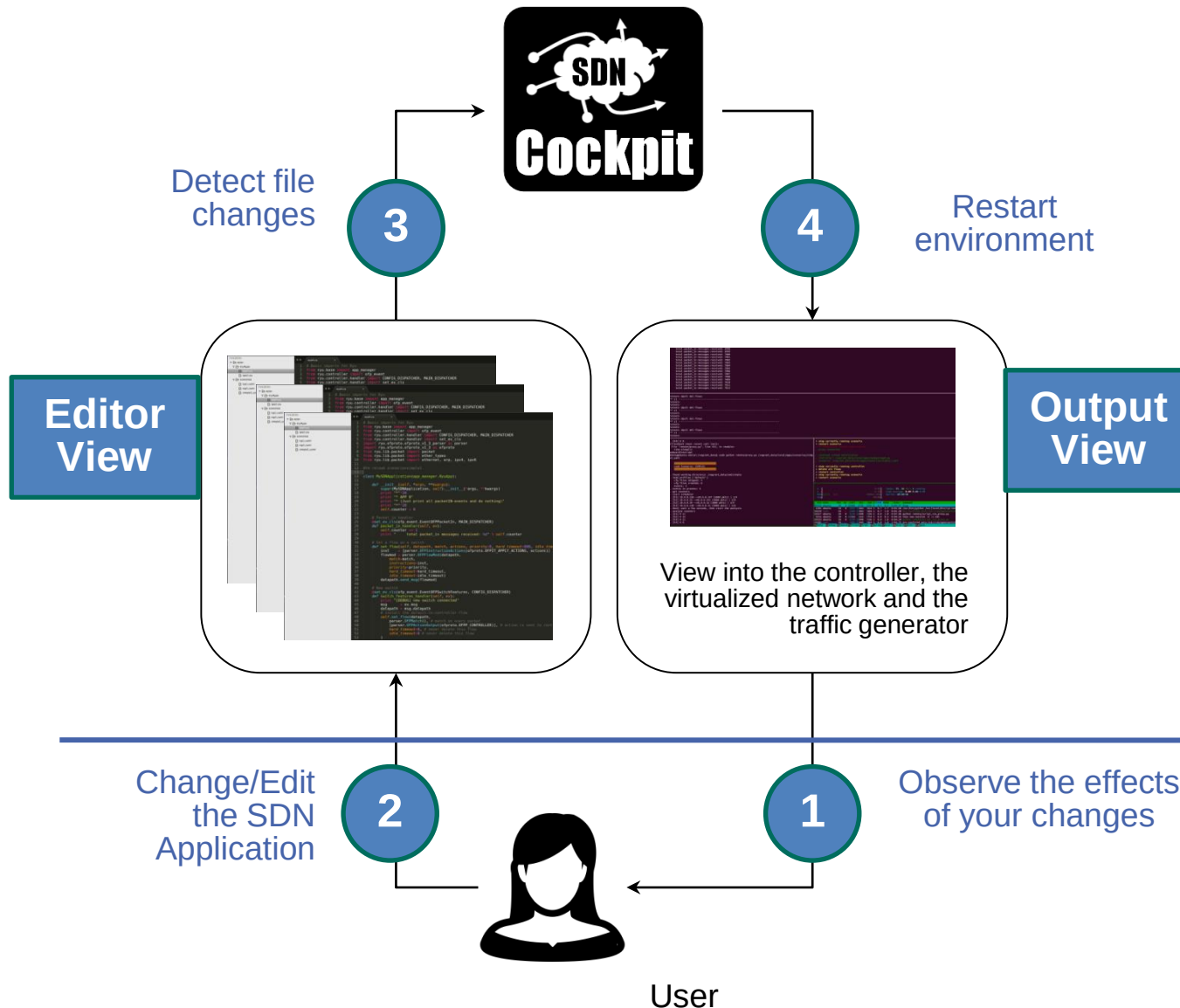
- Idea
 - Collect hands-on experience with SDN + OpenFlow
 - Get familiar with the important fundamentals of SDN application development
 - Based on the **SDN Cockpit tool** that we also used for the Telematik lecture
- Individual work
- Consists of 3(+1) tasks covering
 - Fundamentals (Priorities, Timers, Reactive+Proactive, ...)
 - Switching
 - Routing
 - Monitoring
 - Multiple flow tables & group tables...

Block1: SDN Cockpit

- A Mininet-based Framework for Fast SDN Application Development
- Available at <https://gitlab.kit.edu/kittm/telematics/sdncockpit>
- **Easy to use**
 - Start without training
 - Configuration of the tools etc is done completely automatic
 - Simple Web Interface
 - No unnecessary hurdles
- **Integrated scenario / traffic**
 - Preconfigured scenarios that are „replayed“ automatically
 - This cannot be done with „pure“ mininet
- **Automatic evaluation**
 - Does the application run properly?
 - How good / efficient is my application?
 - Real-time feedback during execution



Block1: SDN Cockpit



Block1: SDN Cockpit



Run Scenarios:

- Mininet
- Openvswitch
- Ryu SDN Controller

File Sync

Output

Choose Scenario

Editor

Change/Edit:
SDN Application

Scenario
View

Log
Output

Test
Output

Choose SDN App

Execute Actions:
Start/Stop Simulation
Automatic Evaluation
PingAll
DumpFlows

Block2: Reverse-engineering + Evaluation

- Idea
 - Move away from SDN Cockpit
 - Apply the OpenFlow toolbox to a „real“ use case
 - Select one interesting SDN application that was presented in a recent research paper
 - **Implement** (maybe in a simplified form)
 - **Evaluate** some key performance indicators
 - **Compare** with results published in the paper
- Teamwork (in groups of 3)
- Free choice of topics (we will prepare a list with proposals)
 - Security
 - Load Balancing
 - Performance Optimization
 - ...

Agenda for Today

- What is this practical course about?
- **Time schedule**
- Overview of graded documents/activities
- Next Steps

Schedule (preliminary)

KW	Date	Meeting	Homework / Todos
17	2026-04-20	KickOff + Intro (SDN & SDNCockpit)	Confirm Participation (Register in CAS) + Preparations (1 week)
18	2026-04-29	Q & A Preparations / Task 0 optional	Task 1/3: Deadline 2026-05-06T14:45
19	2026-05-06	Discussion Task 1	Task 2/3: Deadline 2026-05-13T14:45
20	2026-05-13	Discussion Task 2 + Intro Block 2	Task 3/3: Deadline 2026-06-03T14:45 (2 weeks)
21	2026-05-20	- no meeting -	Skim Papers
22	2026-05-27	- no meeting (vacation) -	
23	2026-06-03	Discussion Task 3	Select Papers
24	2026-06-10	Discussion optional Task 4	
25	2026-06-17	Block 2: Intro Meeting (presentations)	
26	2026-06-24	- buffer / no meeting -	
27	2026-07-01	Block 2: Interim Meeting (with presentations)	Work on Reproduction (7 weeks)
28	2026-07-08	- buffer / no meeting -	
29	2026-07-15	Block 2: Individual Meetings	
30	2026-07-22	- buffer / no meeting -	
31	2026-07-29	Final Meeting with presentation	Prepare final Report

Agenda for Today

- What is this practical course about?
- Time schedule
- **Overview of graded documents/activities**
- Next Steps

Overview of graded documents/activities

- So what has to be done during the course?
 - Participation in our **weekly meetings**
 - **Block 1:**
 - Submit solutions for the **three tasks** (commit by deadline)
 - Give one **5-10 minute presentation** on your solution
 - **Block 2:**
 - Provide solution for the **project/reengineering task** (teamwork)
 - Give one **20 minute presentation** (teamwork)
 - Provide a brief **evaluation report** about the results (teamwork)

Agenda for Today

- What is this practical course about?
- Time schedule
- Overview of graded documents/activities
- **Next Steps**

Next Steps

- If you want to participate:
 - Make sure you signed the sheet
 - [Register online](#) (student portal)
- **Matrix Channel – Communication**
 - Login once now so we can invite you!
 - Discussions, Technical problems, ...
- **Gitlab – Course Material, Submissions**
 - Login once now so we can invite you!
 - Repository for individual and group submissions
- [TODO 1: Task 0 – Setup Environment \(1 week\)](#)
- (TODO 2: Task 1/3 (2 weeks))



Matrix



Gitlab

Any Questions?



Contacts:

- Michael König <m.koenig@kit.edu>
- Paul Seehofer <paul.seehofer@kit.edu>